

Answer all questions.

1. (a) The following diagrams show the life cycle of two different mass stars.
Diagram 1 shows how a star that is similar in mass to our Sun changes with time.
Diagram 2 shows the changes for a star that is 8 times more massive than our Sun.

Complete the labelling on diagrams **1** and **2**, using words or phrases from the box below.
 Words or phrases can be used once, more than once or not at all. [4]

asteroid neutron star red giant supergiant white dwarf

Diagram 1 – Star of mass similar to our Sun

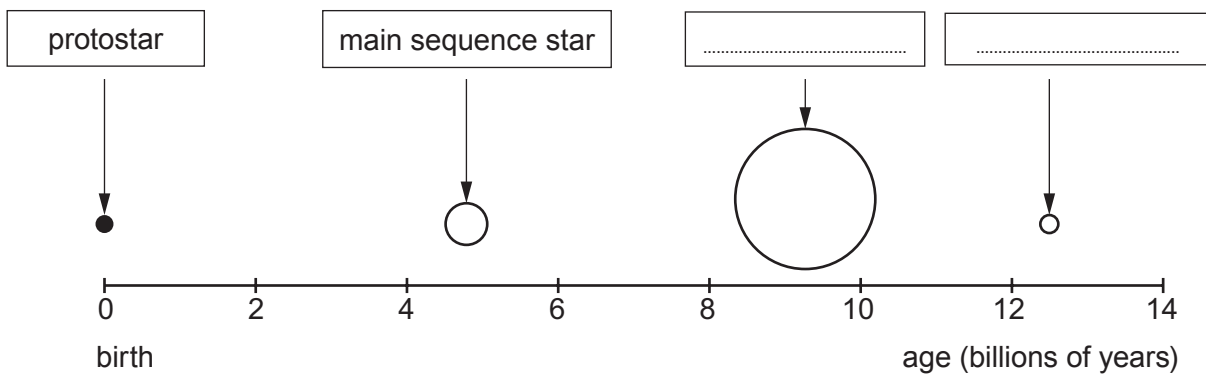
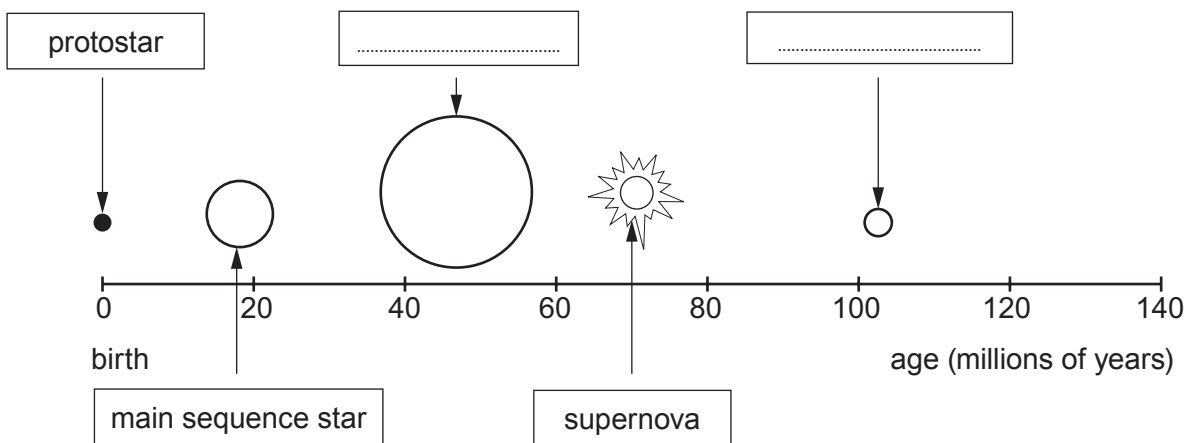


Diagram 2 – Star that is 8 times more massive than our Sun



(b) **Underline** the word or phrase in the brackets which correctly completes each sentence. [4]

- (i) A main sequence star is stable because its gravitational force is
(**less than** / **equal to** / **greater than**) the force caused by its radiation pressure.
- (ii) Stars generate their energy by the (**burning** / **fusion** / **fission**) of increasingly
heavier elements.
- (iii) During the final stages in the life cycle of some stars heavy elements are ejected
when the star becomes a (**gaseous giant planet** / **supernova** / **supergiant**).
The collapse of a cloud of gas and dust combined with these heavy elements can
eventually form a (**Solar System** / **galaxy** / **Universe**).

